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Graph Neural Networks for Detecting Signs of Depression from Social Media Text

Bachelor's Project Thesis

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Abstract: Social media has grown into a root place for exchanging ideas and facilitating communication. Particularly, people with mental health conditions like depression use social media to communicate their feelings and look for help. This trend prompts research into the application of natural language processing (NLP) methods and an assessment of graph neural network's (GNN's) efficacy. In order to predict the severity of depression using textual data, we present three graph construction approaches in this paper: dependency parsed, context window and extended context window. According to our findings, the dependency GNN with BERT embeddings outperforms the baselines, including ensemble Transformer techniques, and exhibits state-of-the-art performance on the Reddit posts dataset. Additionally, we demonstrated how ordinal regression enhances model predictions in extreme classes—an important area for clinical applications. We publish all source code on GitHub*.

1 Introduction

According to World Health Organization (2017), depression is a common mental disorder that affects over 300 million people globally and is a major contributor to the burden of mental health-related diseases (Herrman et al., 2019). A wide range of symptoms are commonly observed in depressed individuals, including persistent melancholy and unusual thoughts, feelings, emotions, and actions. Depression that goes undiagnosed and untreated can have serious repercussions, including a higher risk of suicide. Studies indicate that more than 5% of depressed teenagers may end their lives, underscoring the significance of prompt identification and assistance (Patel et al., 2016). Depression can lead to social disengagement and isolation, compounding the challenges faced by people suffering (Luo & Hancock, 2020; Naslund et al., 2014).

There are less than one mental health professional for every 100,000 individuals across multiple low- and middle-income countries Santomauro et al. (2021). The efficiency of diagnosing and treating depression is seriously hampered by this deficit, depriving a large number of people of the care and assistance they require.

Depression, despite its devastating repercussions, is a curable condition (Centers for Disease Control and Prevention (CDC), 2010). Early diagnosis and appropriate treatment can significantly improve individual outcomes by reducing symptoms and improving quality of life. To lessen depression's long-term effects, prompt intervention is crucial (Harrington & Clark, 1998).

Advances in machine learning have opened up new avenues for early, non-invasive diagnosis of depression. It has shown that using natural language processing (NLP) analysis on social media data is a useful method for diagnosing depression early on (Lorenc et al., 2022; Coppersmith et al., 2014).

Early NLP text analysis methods relied on vocabulary-based approaches, where texts were evaluated by adding up the scores from sentiment lexicons that have been precomputed. These methods, however, lacked contextual awareness and domain-specific adaptation. Then came the bag of words (BOW) models, which represented texts as unordered collections of words and neglected language's sequential feature. These techniques did not capture semantic subtleties; instead, they used a variety of classifiers to predict outcomes based on

*https://github.com/nomomon/bachelor-thesis word frequencies.

NLP advanced significantly once neural networks were popularized. Word embeddings, which represent words in a vector space and capture the semantic connections between them, are used in these models. Convolutional Neural Networks (CNNs) extracted local features by using convolutional layers on word embeddings to classify texts and enhance the model's understanding of context (Kim, 2014). By consecutively processing word inputs and capturing temporal text properties, Recurrent Neural Networks (RNNs)—including Long-Short Term Memory (LSTM) and Gated Recurrent Unit (GRU) variants—further enhanced text analysis (Sherstinsky, 2018). However, both CNNs and RNNs have limitations in terms of computing efficiency and contextual integration over bigger texts Sherstinsky (2018).

Recent work has focused on Graph Neural Networks (GNNs), which represent texts through nodes and edges in graph structures, to capture complex word inter-relationships inside texts. Early research has employed GNNs for tasks related to broad text categorization and semantic analysis (Li & Li, 2022; Huang et al., 2019; Liao et al., 2021). However, its application in emotional and depression text analysis remains underexplored. With the use of sophisticated embedding techniques like Word2Vec and BERT for word vector representation, known graph construction methods and ordinal regression for optimization, this paper aims to evaluate the effectiveness of GNNs in diagnosing different severity levels of depression from textual data.

Given the context and the need for effective methods in depression detection from textual data, this study addressed the following research questions:

- **RQ1**: Are graph-based methods more effective than baseline methods in classifying depression severity levels?
- **RQ2**: Does the inclusion of ordinal regression improve the learning outcomes of the model?
- **RQ3**: Which embedding technique is best suited to the task at hand?

The paper is organized as follows: Section 2 describes the relevant research in the field of NLP-based depression detection. Section 3 explains in depth the approach, model architecture, and graph input construction. Section 4 presents the experimental setup, including the dataset and baseline techniques used for comparison. Section 5 reports the findings and discussion. Lastly, Section 6 concludes the paper and future study directions are outlined.

2 Related Work

Given the complicacy and subtleness of mental health disorders, identifying them using NLP and machine learning approaches is more difficult than for expert psychologists. Yet if these methods are effectively applied, they can identify signs of mental health disorders at early stages, enabling patients to be directed to a specialist for a more thorough diagnosis. There are several privacy and ethical concerns that limit the availability of clinically verified datasets for mental diseases, in contrast to traditional machine learning tasks that benefit from huge, high-quality datasets with gold-standard diagnoses. Because of this, a lot of researchers concentrate on developing models for identifying mental health issues using either their own datasets or datasets that are made available to the public.

Previous methods, including Coppersmith et al. (2014), gathered tweets from users who publicly shared their diagnosis of mental illness on Twitter. By examining people's language patterns, the authors searched for linguistic patterns associated with mental health conditions. They found that particular word choices and phrases may indicate mental health problems. In a subsequent study, Coppersmith et al. (2016) used logistic regression with character n-gram features to identify suicidal intentions in tweets. Although, these techniques were effective, they fall short in fully capturing the complexity of the language used in mental health illnesses.

With the introduction of neural networks and the Transformer architecture (Vaswani et al., 2017), models captured better semantics and context. For instance, Sampath & Durairaj (2022) applied traditional learning algorithms, finding that a model using a Word2Vec (Mikolov et al., 2013) embeddings and Random Forest classifier performed best. The authors published their expert-labeled dataset with three classes of depression (not-, moderately-

, and severely-depressed) from social media postings. Later, this dataset was used in Sampath et al. (2022)'s DepSign-LT-EDI@ACL-2022 shared task, which aimed to classify social media text into respective depression levels. The results showed, out of the 31 participating teams the Transformer-based models scored better than other modes. The winning team developed an ensemble model of two fine-tuned RoBERTas on the provided and extra dataset (Poświata & Perełkiewicz, 2022).

Similarly, Tavchioski et al. (2023) built and fine-tuned several large pre-trained language model-based classifiers, including BERT, RoBERTa, BERTweet, and mentalBERT, for detecting depression from social media posts. Their analysis demonstrated that Transformer ensembles outperform single Transformer-based classifiers. These studies showed the effectiveness of embedding methods like Word2Vec and Transformer models in capturing semantic and contextual meaning, which is why we also used them as text-to-vector encoders.

Very few studies have investigated predicting the severity of depression based on ordinal regression. Jayawardena et al. (2018) highlights the importance of using ordinal regression algorithms for predicting depression severity and compares these methods with multiclass classification and regression using a support vector framework. They also introduce a novel approach to calculate rank boundaries for ranking algorithms and demonstrate its effectiveness through experiments on synthetic data and the DAIC-WOZ depressed speech database. Jacobucci et al. (2021) introduced ordinal regression as a statistical method specifically suited to examine distinct suicide outcomes, ranging from suicidal ideation to suicidal planning. They demonstrated the utility of this method using data on depression and nonsuicidal self-injury, highlighting its potential to identify unique risk factors for each suicide outcome and to inform theoretical models of suicide and risk prediction. Recent deep learning research lacked in applying ordinal regression, Shi et al. (2021) proposed a new method for rank-consistent ordinal regression for deep neural networks, which is architecture agnostic and they have shown that on various datasets the method utilizes the ordinal information.

GNNs have drawn interest recently in NLP tasks including text categorization (Yao et al., 2019; Huang et al., 2019). Text_GCN from Yao et al. (2019) was one of the earliest methods to create a single graph for the full corpus, which resulted in significant memory usage and no support for online testing. Huang et al. (2019) proposed another GNN-based text model, where they constructed a text-level graph that connected word nodes inside a short fixed window. However, this technique only takes into account adjacent nodes and ignores dependencies between faraway nodes. A unique multi-level GNN for text sentiment analysis was later proposed by Liao et al. (2021); this network is similar to the CNN network proposed by Kim (2014). Different size node connection windows were used to capture both local and global features. To improve the feature aggregation of each word node in the graph, they also added the attention mechanism (Veličković et al., 2017). Their experiments showed that this model performed better in text sentiment analysis tasks than other existing approaches. Following a similar approach, Li & Li (2022) developed a sentiment classification model based on a GNN to handle the non-standardized and colloquial form of Weibo comments. Unlike previous studies, they used dependency syntax trees to construct semantic graphs for short texts and designed a spatial domain graph filter for feature extraction while employing LSTM as a state updater to filter node noise, achieving superior performance with a 95.22% F1 score.

Although there has been substantial advancement in the use of GNNs for text classification tasks in prior research, there has been limited investigation into the use of GNNs for depression detection. No study has paired GNNs with ordinal regression for this purpose, despite the fact that it has been demonstrated that ordinal regression is useful in predicting the severity of depression. This work uses these methods and three graph construction approaches for predicting depression severity from textual data, namely dependency GNN, extended context window, and context window, in an attempt to fill in these gaps. We assess the effectiveness of these techniques in comparison to conventional machine learning models and look into how ordinal regression affects the learning results. Additionally, we assessed the performance of different embedding techniques, including Word2Vec and BERT, in the context of depression severity classification.

3 Methods

In this section, we present the methods from text embeddings and graph construction methods, shown in Figure 3.1, to the model architecture, shown in Figure 3.2.

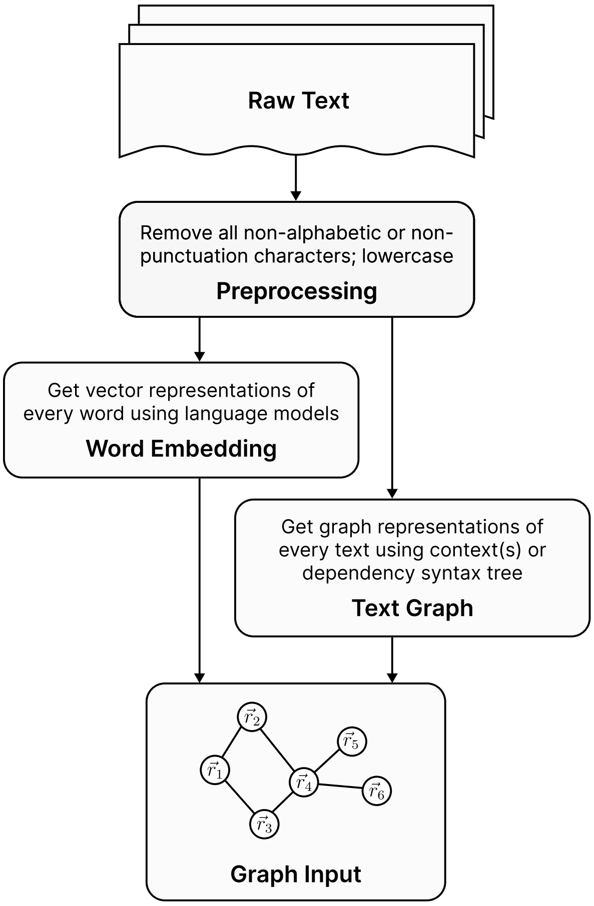

Figure 3.1: Text processing flowchart, to create the graph input. Raw text is preprocessed, word-embedded to get vector representations, and finally used to generate text graphs. The graph edges and node vectors are the graph input.

The dataset is a collection of texts, social media posts, written by individuals with labeled levels of depression severity. The text data is converted into a graph structure by mapping every word in a text to a node and the relationships between these words to edges in a graph. This text's graph can be expressed as $G = (V, E)$, where V stands for the set of words' nodes and E for the set of edges that describe the words' relationships. Each node $v \in V$ has an associated vector representation $\vec{r}_v$ initialized with word embeddings of dimension d_0 produced using a pre-trained language model, which is discussed more thoroughly in Section 3.1.The production of graph edges, or relationships between words is defined and explored later in Section 3.2.

These embeddings and graph edges are passed through a graph attention block, which computes the attention weights for every node's neighbours, aggregates them and passes through a fully connected layer. This operation is executed three times, inspired by prior works, to get the final node embeddings. The vectors are aggregated using global max and average pooling and forwarded through two fully connected layers to produce the output. The output is post-processed using the ordinal regression decoder, giving the final predicted labels. Finally, the model is trained using an ordinal regression loss. The details of the methods are discussed in the following sections.

3.1 Embedding Techniques

The text was transformed into vector space using Word2Vec (Mikolov et al., 2013) and BERT (Devlin et al., 2019) embedding models. Word2Vec captures semantic associations based on context from its training set, whereas BERT, a bidirectional Transformer, uses the surrounding context of a word to provide differential embeddings.

Word2Vec embeddings were obtained from a pre-trained Word2Vec model, `google-news-300`. A random vector from a normal distribution was sampled as a substitution for unknown words, according to the methodology that has been proven successful in earlier research (Kim, 2014). The BERT embeddings were retrieved from the pre-trained BERT model, notably the Hugging Face's `BERT-base-uncased`. Word-by-word tokenization was applied to the words, and `[UNK]` was used to tokenize unknown ones.

3.2 Building Text Graph

We explored the following graph construction methods to represent text data in a graph format:

- *Context Window*: Words within a certain radius are connected.

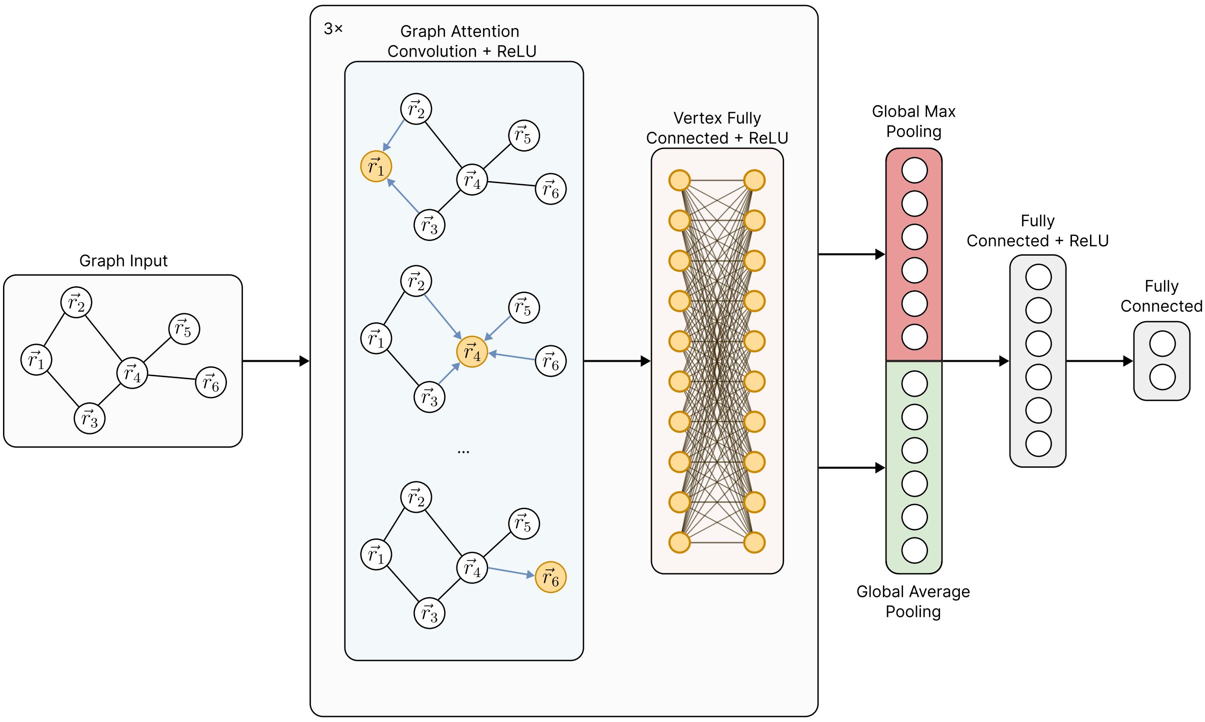

Figure 3.2: The architecture of the GNN model used, it has 3 repeating blocks of graph attention convolutional layers followed by fully connected layers for vertex vectors. After that, the vertex vectors are aggregated with global max and average pooling and concatenated. It is passed through another fully connected layer with ReLU activation and finished with an output layer of size 2.

- *Extended Context Window*: Same as context window, but the radius increases every layer of the GNN.
- *Dependency Graph*: Words are connected if they are connected in the dependency graph.

These techniques capture various aspects of text relations such as local context, global context and syntactic relationships respectively.

3.2.1 Context Window Graph

This text graph was used in Huang et al. (2019). Let, each word v_i in the text be connected to every other word v_j within a predefined window size k. The edge set E has the following mathematical definition:

$$E = \{(v_i, v_j) \mid |i - j| \leq k,\ v_i, v_j \in V\}, \quad (3.1)$$

where $|i - j|$ stands for the positional difference between words in the text.

3.2.2 Extended Context Window Graph

This method, inspired by the work of Liao et al. (2021), aims to progressively integrate more contextual information into the node representations as the depth increases. It extends the idea of the previously discussed context window graph by increasing the window size k at each depth level of the GNN, ultimately ending with a fully connected graph in the final layer. For each layer l of the GNN, the edge set E_l is defined as:

$$E_l = \{(v_i, v_j) \mid |i - j| \leq k_l,\ v_i, v_j \in V\}, \quad (3.2)$$

where k_l is the context window size at layer l. Initially, k_1 is small to capture local context, and it increases with depth to incorporate broader context, culminating in $k_L = |V| - 1$ in the last layer $L = 3$, where every node is connected to every other node.

3.2.3 Dependency Graph

This text graph was used in Li & Li (2022). The dependency graph leverages syntactic structures derived from a semantic dependency parser, connecting words that have direct grammatical relations in the text. The edge set E is defined as

$$E = \{(v_i, v_j) \mid v_i \text{ is related to } v_j \text{ in the dependency tree}\}. \quad (3.3)$$

The dependency graphs were obtained from the spaCy's large English dependency parser (Honnibal & Montani, 2017).

3.3 Graph Attention Block

The message-passing mechanism uses graph attention convolutions (Veličković et al., 2017). Equation 3.4 is used to calculate the correlation coefficient between neighboring nodes. Equation 3.5 is a softmax of the correlations for every neighbor, yielding the attention weights. For layer $l \in \{0, 1, 2\}$, the mechanism is defined by

$$c_{ij} = \text{LeakyReLU}\left(\mathbf{w}_l(W_l r_i^{(l)} \oplus W_l r_j^{(l)})\right) \quad (3.4)$$

and

$$a_{ij} = \frac{\exp(c_{ij})}{\sum_{k \in N(i)} \exp(c_{ik})} \quad (3.5)$$

where the $\oplus$ represents the concatenation operation, $\mathbf{w}_l \in R^{2d_{l+1}}$, $W_l \in R^{d_l \times d_{l+1}}$ are trainable parameters, LeakyReLU is a kind of non-linear activation function (Maas, 2013), and $N(i)$ is the set of direct neighbors of node i. Then, the representation of adjacent nodes is weighted and aggregated, transforming into a new node representation $\hat{r}_i^{(l+1)}$ of dimension d_{l+1}, as follows:

$$\hat{r}_i^{(l+1)} = f\left(\sum_{j \in N(i)} a_{ij} W r_j^{(l)}\right) \quad (3.6)$$

This middle representation is passed through a vertex level fully connected layer to get the final representation of the words $r_i^{(l+1)}$,

$$r_i^{(l+1)} = f\left(W_{fc}^{(l)} \hat{r}_i^{(l+1)} + b_{fc}^{(l)}\right) \quad (3.7)$$

Here, $W_{fc}^{(l)} \in R^{d_l \times d_{l+1}}$, $b_{fc}^{(l)} \in R^{d_{l+1}}$ are trainable parameters and f is a non-linear activation function. The process is repeated for each layer $l \in \{0, 1, 2\}$, as in the architectures of Liao et al. (2021); Li & Li (2022).

3.4 Aggregation & Output

After the graph attention blocks, the new vector embeddings $r_i^{(3)}$ are aggregated using global max pooling and global average pooling, and concatenated. The global average pooling is computed as equation 3.8 and global max pooling is computed as equation 3.9.

$$r_{\text{avg}} = \frac{1}{N} \sum_{1}^{N} r_i^{(3)} \quad (3.8)$$

$$r_{\max} = \left[\max_i r_{i,0}^{(3)}, \max_i r_{i,1}^{(3)}, \ldots, \max_i r_{i,d_3}^{(3)}\right]^T \quad (3.9)$$

These feature vectors are concatenated to get the final representation of the text $r_{\text{text}} = r_{\text{avg}} \oplus r_{\max}$. These features are passed through a fully connected layer with a non-linear activation.

$$x_{\text{hidden}} = f\left(W r_{\text{text}} + b\right) \quad (3.10)$$

$$x_{\text{out}} = \text{softmax}\left(W x_{\text{hidden}} + b\right) \quad (3.11)$$

Finally, they are again passed through a fully connected layer to produce the output logits. As proposed by Shi et al. (2021), to achieve ordinality, the output logits correspond to the probability of each severity class being greater than a certain threshold. Thus, providing an ordinal relationship between classes.

To obtain the severity level $q \in \{1 = \text{not}, 2 = \text{moderate}, 3 = \text{severe}\}$ of the i-th of a sample, we threshold the predicted probabilities corresponding to the two binary tasks and sum the binary labels as follows:

$$q^{[i]} = 1 + \sum_{j=1}^{2} \mathbb{1}\left(\hat{P}\left(y^{[i]} > r_j\right) > 0.5\right),$$

where the predicted rank is $q^{[i]}$. See Section 4.1 for rank descriptions.

This training scheme uses conditional training sets to obtain the unconditional rank probabilities through applying the chain rule for conditional probability distributions.

3.5 Loss Function

To exploit the ordinal nature of the dataset labels, we set the goal of training to minimize the categorical ordinal regression network (CORN) loss from Shi et al. (2021), between the ground truth label and the predicted label as follows:

Let $x_{\text{out},j}$ denote the predicted value of the j-th node in the output layer of the network (Eq. 3.11), and let $|S_j|$ denote the size of the j-th conditional training set. To train a CORN neural network using backpropagation, we minimize the following loss function:

$$L(\mathbf{X}, \mathbf{y}) = \\ -\frac{1}{\sum_{j=1}^{K-1} |S_j|} \sum_{j=1}^{K-1} \sum_{i=1}^{|S_j|} \left[\log\left(x_{\text{out},j}(\mathbf{x}^{[i]})\right) \cdot \mathbb{1}\left\{y^{[i]} > r_j\right\} \right. \\ \left. + \log\left(1 - x_{\text{out},j}\left(\mathbf{x}^{[i]}\right)\right) \cdot \mathbb{1}\left\{y^{[i]} \leq r_j\right\} \right], \quad (3.12)$$

It must be noted that $x_{\text{out},j}(\mathbf{x}^{[i]})$, $\mathbf{x}^{[i]}$ represents the i-th training example in S_j. To simplify the notation, we omit an additional index j to distinguish between $\mathbf{x}^{[i]}$ in different conditional training sets.

4 Experimental Setup

4.1 Dataset & Preprocessing

The dataset used in this study was sourced from Reddit, specifically from subreddits `r/mentalhealth`, `r/depression`, `r/loneliness`, `r/stress`, and `r/anxiety` (Sampath & Durairaj, 2022). The dataset contains posts of users' feelings and thoughts, which were annotated by mental health professionals into three categories of depression severity:

1. **Not depressed**: Posts with no signs of depression, containing irrelevant content or offering help and motivation to individuals with depression.
2. **Moderate**: Posts with moderate signs of depression, indicating a change in feelings but showing signs of improvement and hope.
3. **Severe**: Posts with severe signs of depression, often related to serious suicidal thoughts, disorder conditions, or past experiences.

This dataset in total has 10251 samples, of which 6006 are training, 1000 are validation and 3245 are testing samples.

Prior to the split, issues such as data leakage and duplicates were identified and removed. The class distribution in the dataset is imbalanced, which has been addressed during model training through techniques such as weighted sampling to ensure an equal representation of each class. All texts were standardized by removing non-alphabetic and non-punctuation characters, converting to lowercase.

This dataset was also featured in the DepSign-LT-EDI@ACL-2022 competition Sampath et al. (2022), providing a comparative framework for our analysis. The top five methods from the competition are discussed in Section 4.4.1.

4.2 Evaluation Metrics

Model performance was assessed on the test set with the primary metric being the macro F1 score. In addition, we also measured the accuracy, macro precision, and macro recall. Furthermore, confusion matrices were created in some cases for more granular insights into the model's performance.

Due to the dataset's class imbalance, the macro F1 score was chosen as the key measure to ensure that the model's performance is evaluated evenly across all classes. The macro F1 score is calculated as the average of the F1 scores for each class.

4.3 Hyperparameters and Training

To optimize the models' performance, Bayesian hyperparameter optimization Masum et al. (2021) was employed due to its efficiency in requiring fewer iterations compared to exhaustive search methods. The parameter search included the number of attention heads, embedding size in the attention block, dense neurons in the output, weight decay, batch size, dropout rate, learning rate, and context window sizes (k_i for specific layers). Liao et al. (2021) have shown the significance of selecting proper context sizes on the performance of the model and that the range of exploration should be from 1 to 8. The final hyperparameters as well as the parameter search space for each model, using BERT and Word2Vec embeddings, are detailed in Table 4.1.

For all models we used ReLU (Agarap, 2018) as the non-linear activation function, and the Adam optimizer (Kingma & Ba, 2015). The models were trained for 100 epochs with early stopping based on the validation loss. Early stopping was triggered when the validation loss did not improve or increase for more than three consecutive steps. The models were implemented using PyTorch (Paszke et al., 2019) and trained on an NVIDIA Tesla V100 GPU from Google (2024).

4.4 Baselines

We developed a number of baseline techniques in order to assess our proposed models. A second dependency GNN has been trained using categorical cross entropy and it served as a baseline for the loss function. We tuned hyperparameters for each baseline using grid search, optimizing for the best performance on the validation set. The following baseline models were used:

Logistic Regression Two techniques, Bag of Words (BOW) and Average Vector Embedding (AVE), use distinct embeddings, and the logistic regression model is used as the classifier in both. The BOW model counts the frequency of each word in text, regardless of word order, and converts that data into fixed-length vectors. Conversely, the AVE model uses vector embeddings, like Word2Vec and BERT, by encoding each text as the mean vector of its individual word embeddings.

Convolutional Neural Network (CNN) We have adapted the architecture from Kim (2014) to create our CNN baseline. It is implemented for Word2Vec and BERT embeddings. In order to extract distinct n-gram features from sentences, the CNN employs a variety of filter sizes. The pooling layer captures the most significant features. After that, a fully connected layer with ReLU activation and a dropout of 0.5 is applied to these features in order to prevent overfitting, and a softmax layer for class prediction is the result.

Long Short-Term Memory (LSTM) Our implementation includes an embedding layer followed by LSTM cells. The output from the LSTM cells passes through a fully connected layer and then to a softmax layer for final classification. This model is employed with both Word2Vec and BERT embeddings. A dropout rate of 0.5 is used before the LSTM layer to enhance model generalization.

4.4.1 Competition Baselines

The dataset used in this study was also featured in the DepSign-LT-EDI@ACL-2022 competition (Sampath et al., 2022). For comparison, we used the top 5 teams from the competition according to the macro F1 metric. The top 5 teams are as follows:

- OPI (Poświata & Perełkiewicz, 2022): Used an ensemble of predictions from a fine-tuned RoBERTa large model on an augmented dataset and a fine-tuned RoBERTa large model on a parsed Reddit Mental Health dataset.

- NYCU_TWD (Wang et al., 2022): Used an ensemble of machine learning-based models, pretrained language models, and pretrained language models with supervised contrastive learning, utilizing VAD scores and a power weighted sum technique.

- ARGUABLY (Sampath et al., 2022): Used an ensemble of XLNET, BERT, and RoBERTa models, and a fine-tuned RoBERTa model on the dataset.

- BERT 4EVER (Lin et al., 2022): Used a BERT model with sentiment embedding and adversarial training, and a T5-base model with prompt-learning and back-translation.

- KADO (Janatdoust et al., 2022): Used a BERT based Hybrid Transformer model for classification.

These teams use a mix of ensemble methods and single models, with some using fine-tuning and others using pre-trained models. These models are a useful comparison to our suggested models because they show a wide range of what is achievable in the field.

GNN Model	Hyperparameter	BERT value	Word2Vec value	Search space
Context Window	N attention heads	5	1	$\{1, 2, \ldots, 5\}$
	Weight decay	10^{-2}	10^{-5}	$\{10^i \| i \in \{-5, -4, \ldots, 0\}\}$
	Dense neurons	64	128	$\{32, 64, 128, 256\}$
	Batch size	32	128	$\{32, 64, 128, 256\}$
	Dropout	0.3	0.5	$\{0.1, 0.2, \ldots, 0.5\}$
	Embedding size	64	32	$\{32, 64, 128, 256\}$
	Learning rate	10^{-3}	10^{-3}	$\{10^i, i \in \{-4, -3, \ldots, 0\}\}$
	k	5	5	$\{1, 2, \ldots, 8\}$
Extended Context Window	N attention heads	5	5	$\{1, 2, \ldots, 5\}$
	Weight decay	10^{-3}	10^{-3}	$\{10^i \| i \in \{-5, -4, \ldots, 0\}\}$
	Dense neurons	128	128	$\{32, 64, 128, 256\}$
	Batch size	64	128	$\{32, 64, 128, 256\}$
	Dropout	0.1	0.1	$\{0.1, 0.2, \ldots, 0.5\}$
	Embedding size	32	64	$\{32, 64, 128, 256\}$
	Learning rate	10^{-3}	10^{-4}	$\{10^i \| i \in \{-4, -3, \ldots, 0\}\}$
	k_1	2	2	$\{1, 2, \ldots, 7\}$
	k_2 ($k_2 > k_1$)	4	3	$\{2, 3, \ldots, 8\}$
Dependency	N attention heads	5	5	$\{1, 2, \ldots, 5\}$
	Weight decay	10^{-4}	10^{-4}	$\{10^i \| i \in \{-5, -4, \ldots, 0\}\}$
	Dense neurons	64	64	$\{32, 64, 128, 256\}$
	Batch size	128	128	$\{32, 64, 128, 256\}$
	Dropout	0.1	0.1	$\{0.1, 0.2, \ldots, 0.5\}$
	Embedding size	64	64	$\{32, 64, 128, 256\}$
	Learning rate	10^{-4}	10^{-4}	$\{10^i \| i \in \{-4, -3, \ldots, 0\}\}$

Table 4.1: The best hyperparameters for GNN models trained using BERT and Word2Vec embeddings found using Bayesian parameter optimization. The search space for each hyperparameter is also shown.

5 Results and Discussion

The results for the seven models and their different embedding variants (thirteen in total) are shown in Table 5.1. BOW serves as a strong baseline, outperforming all other baselines except CNNs with BERT embeddings. CNN variants have better performance than AVE and LSTMs with Word2Vec and BERT embeddings. LSTMs have consistently performed better than AVE. The AVE model didn't outperform other models; however, while using BERT embeddings, it did better than LSTM with Word2Vec. Nevertheless, AVE with Word2Vec achieved the highest accuracy. Overall, methods that used BERT embeddings outperformed their counterparts that used Word2Vec.

The GNN models had the same architectures but used different text-to-graph construction methods: context window, extended context window, and dependency tree. The results show that GNN-based methods can effectively improve the performance of text depression detection. The context window and extended context window GNNs with Word2Vec embeddings outperformed all baselines except CNN with BERT embeddings. Notably, both dependency GNN variants outperformed all GNNs and baselines. Furthermore, the BERT dependency GNN had the highest macro precision. Among the GNN models, the effectiveness of graph construction can be ordered as follows: dependency is the best, followed by context window, and then extended context window. The results of the CNNs and the extended context window and context window GNNs are similar, but the GNN variants did slightly better.

Table 5.2 displays the comparison results for the dependency GNN variants trained with CORN and cross-entropy losses. In terms of accuracy, macro precision, and macro F1-score, the CORN loss function did better than the cross-entropy loss function. The findings show that the ordinal character of depression severity levels is better captured by the CORN loss function, which improves classification performance. Figure 5.1 displays the confusion matrices of the dependency GNN with BERT embeddings trained with various loss functions. In clinical applications, it is desirable to eliminate confusion between extreme classes, and this was achieved with the CORN loss function.

Table 5.3 displays the comparison between the performance of our top model, dependency GNN with BERT embeddings, and that of the DepSign-LT-EDI@ACL-2022 competition competitors. The results show that our model has the highest macro F1-score on the test set. The suggested model outperforms the competitors' methods, which included ensemble models, BERT versions, and fine-tuned models, in the classification of depression severity levels, indicating that dependency GNN with BERT embeddings is a promising method for depression classification tasks.

5.1 Effectiveness of Graph-Based Methods in Depression Classification

Results show that, in fact, graph-based techniques outperform baseline techniques in the classification of depression severity levels. The dependency graph-based representation consistently performs better than other approaches among the various graph creation techniques that have been investigated. The reason for this better performance is that the model can capture both short- and long-range links between words, which keeps the sentence dependancy semantics and local and global connections in the text intact.

A deeper semantic understanding of the text is made possible by the semantic richness encoded in dependency graphs, which is not fully captured at the embedding level. The improved semantic representation makes it easier to classify depression severity with more accuracy. The dependency graph's capacity to preserve sentence structure enables a deeper understanding of the text, which is critical in the context of depression classification.

It is clear by comparing the context window GNN, extended context window GNN, and CNN models' metrics that their accuracy and macro F1 scores are comparatively close. The comparable topologies and architectures of CNNs and GNNs are responsible for this resemblance. This is because development of GNNs—which apply aggregation on a variety of graph structures—has been greatly inspired by CNNs and their grid structure aggregation.

Particularly, unlike CNNs, which use fixed weights, GNNs use an attention mechanism, allowing for dynamic feature weighting. The model is able to concentrate more on relevant words because to this attention mechanism, which improved performance through more focused learning. Compared to CNNs, GNNs' attention mechanism is more flexible, which gives them a clear advantage when processing complex texts.

The best model for depression classification tasks is dependency GNN with BERT embeddings. Its performance over other graph-based approaches and classic baselines highlights its potential as a promising strategy in this field. The findings imply that the dependence GNN's higher performance is mostly due to the semantic depth and structural integrity it supports.

5.2 Impact of Ordinal Regression

In the context of classification approaches, using the CORN technique instead of the usual cross-entropy loss approach had an impact on the confusion matrix results but did not significantly change overall performance metrics. Less confusion between extreme categories was the effect of the CORN strategy, which takes advantage of the ordinal character of depression severity. This is a desirable outcome in clinical settings. The penalization strategy of CORN, which taxes harsher penalties for errors between non-adjacent classes, is responsible for this decrease in misclassification. This quality is particularly useful where accurate differentiation between mild and severe depression can influence treatment decisions and patient outcomes.

5.3 Advantages of BERT over Word2Vec

The consistent outperformance of BERT embeddings over Word2Vec embeddings in all model ar-

	Method	Embedding	Accuracy	Macro Recall	Macro Precision	**Macro F1-score**
Baselines	BOW	-	0.586	0.506	0.483	0.481
	AVE	Word2Vec	**0.667**	0.538	0.424	0.438
		BERT	0.632	0.507	0.451	0.463
	CNN	Word2Vec	0.609	0.465	0.469	0.467
		BERT	0.616	0.518	0.564	0.534
	LSTM	Word2Vec	0.520	0.448	0.513	0.451
		BERT	0.535	0.459	0.535	0.475
GNN (ours)	Context Window	Word2Vec	0.527	0.462	0.562	0.505
		BERT	0.548	0.476	0.580	0.531
	Extended Context Window	Word2Vec	0.603	0.537	0.482	0.496
		BERT	0.623	**0.554**	0.504	0.523
	Dependency	Word2Vec	0.582	0.494	0.575	0.543
		BERT	0.628	0.523	**0.609**	**0.595**

Table 5.1: Results of the baseline and GNN models and their different embedding variants. The best results are shown in bold.

Method	Embedding	Accuracy	Macro Recall	Macro Precision	**Macro F1-score**
Dependency GNN (CORN)	Word2Vec	0.582	0.494	0.575	0.543
	BERT	**0.628**	0.523	**0.609**	**0.595**
Dependency GNN (Cross-Entropy)	Word2Vec	0.506	**0.592**	0.506	0.487
	BERT	0.413	0.512	0.448	0.413

Table 5.2: Comparison of results of CORN and Cross Entropy loss functions on the GNN variants' performance. The best results are shown in bold.

chitectures is among the most striking findings. The context-aware and bidirectional embeddings that BERT can construct provide a significant advantage over Word2Vec's static and context-independent representations. Given the complexity of sentiment analysis and depression classification tasks, where contextual awareness is crucial, this distinction is especially important. BERT's broad semantic breadth is further enhanced by its thorough pre-training on a variety of textual information which also improves the model's performance.

5.4 Limitations

Even with its strengths, the method has its drawbacks, especially when it comes to the scale and representation of the dataset. The dataset is sufficient for exploratory research, but its lack of full population representation may restrict how broadly the results may be generalized. Data collecting is further complicated by privacy issues and a lack of accessible diagnosed cases on online platforms.

One more drawback is that each person's single data point was used to train the model. This method falls short, in the opinion of clinical psychologists, of ascertaining whether the model is actually generalizable. The mental and behavioral states of humans are complicated and subject to large temporal variations. Relying on a single data point per individual may overlook these variations, leading to an incomplete understanding of the model's effectiveness across different contexts and time periods.

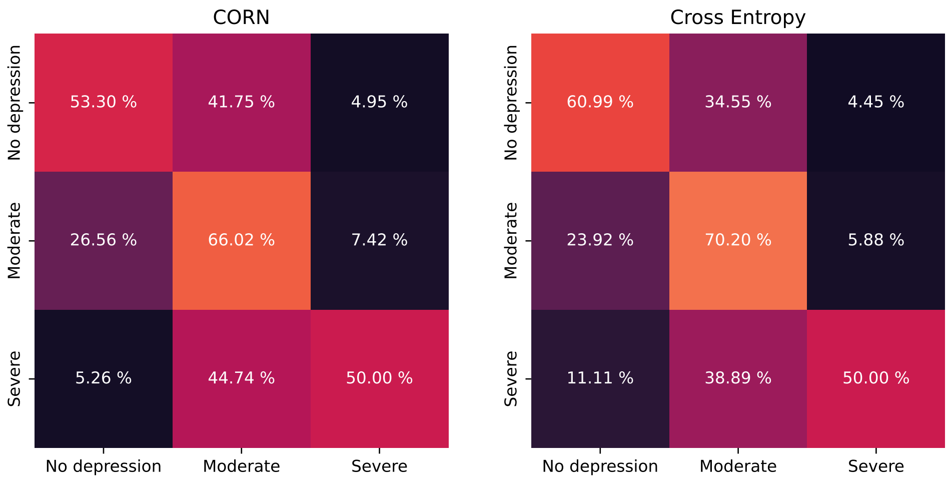

Figure 5.1: Normalized confusion matrices of the GNN with BERT embeddings trained using cross-entropy and CORN loss functions. The CORN loss function reduced the confusion between extreme classes, which is desirable in clinical applications.

Team Name	Accuracy	Macro Recall	Macro Precision	**Macro F1-score**	Rank
OPI	**0.658**	0.565	0.512	0.583	1
NYCU_TWD	0.633	0.539	0.490	0.552	2
ARGUABLY	0.625	0.569	0.479	0.547	3
BERT 4EVER	0.625	**0.581**	0.522	0.543	4
KADO	0.618	0.474	0.498	0.542	5
BERT GNN (ours)	0.628	0.523	**0.609**	**0.595**	-

Table 5.3: Comparison of our best model, dependency GNN with BERT embeddings, to the models from the DepSign-LT-EDI@ACL-2022 competition. Our model has the highest macro F1-score on the test set.

6 Conclusion and Recommendations

In this work, we compared the performance of GNNs against conventional machine learning models and embedding approaches in order to determine how effective they are at predicting the severity of depression from textual data. We did this by utilizing three different graph creation methods. Our findings demonstrate the possibility for increased accuracy and decreased misclassification in the classification of depression severity. Specifically, the dependency GNN with BERT embeddings performs better than traditional techniques. CORN loss did not significantly improve performance over conventional cross-entropy loss, it did improve the outcomes in the confusion of extreme classes, demonstrating its practical value in clinical applications. Our results show that approaches with BERT embeddings generally perform better than those using Word2Vec embeddings. Despite the fact that each individual in our trials received a single data point, the small dataset and underrepresentation limit the generalizability of our results, underscoring the need for more thorough and diverse data gathering techniques.

Further research might investigate the particular patterns in dependency graphs that are associated with depressed symptoms, as this could offer cognitive psychologists and linguists important new insights. Comprehending these trends may result in enhanced diagnostic instruments and remedial measures. Furthermore, it is critical to address privacy issues and guarantee the moral use of personal data for depression assessment in order to advance this field of study without sacrificing the rights of individuals to privacy. Furthermore, by using multi-timepoint data gathering, it may be possible to record the variations in people's mental states over time, leading to a more comprehensive and dynamic knowledge of the severity of depression and how it progresses.

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